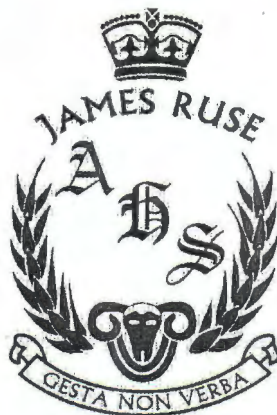


|        |  |
|--------|--|
| Name:  |  |
| Class: |  |



**YEAR 12**

**ASSESSMENT TEST 1**

**TERM 4, 2013**

**MATHEMATICS**

**EXTENSION 1**

*Time Allowed – 90 Minutes*  
*(Plus 5 minutes Reading Time)*

- All questions may be attempted
- Department of Education approved calculators and templates are permitted
- In every Question, show all relevant mathematical reasoning and/or calculations.
- Marks may not be awarded for careless or badly arranged work
- No grid paper is to be used unless provided with the examination paper

**The answers to all questions are to be returned in separate bundles clearly labeled Question 1, Question 2, etc.**

**Each question must show (in the top right hand corner) your Candidate Number.**

**QUESTION 1 (10 Marks)****COMMENCE A NEW PAGE****MARKS**

- (a) Find:  $\int x^2 e^{x^3} dx$  1
- (b) Find:  $\lim_{x \rightarrow 0} \frac{1 - \cos 2x}{x^2}$  3
- (c) Find the exact value of:  $\sin^{-1}\left(-\frac{\sqrt{3}}{2}\right) + \tan^{-1}\left(\tan \frac{7\pi}{8}\right)$  2
- (d) (i) Prove  $\cos(A + B) - \cos(A - B) = -2 \sin A \sin B$  2
- (ii) Hence find the exact value of  $\cos 75^\circ - \cos 15^\circ$  2

**QUESTION 2 (10 Marks)****COMMENCE A NEW PAGE**

- (a) Differentiate  $\cos^{-1}(2x)$ . 1
- (b) Use Simpson's Rule with 3 function values to estimate 2
- $$\int_0^{\frac{1}{2}} \frac{1}{2} \cos^{-1}(1 - 2x) dx$$
- (c) (i) Show that  $\frac{5x^2 + 4}{x^2 + 4} = 5 - \frac{16}{x^2 + 4}$  1
- (ii) Hence, evaluate to 2 decimal places, 2
- $$\int_{-\frac{1}{2}}^0 \frac{5x^2 + 4}{x^2 + 4} dx$$
- (d) The vertices of  $\triangle ABC$  are  $A(4, -3)$ ,  $B(-9, 7)$  and  $C(1, k)$ . If the area of  $\triangle ABC$  is 15 square units, find the value(s) of  $k$ . 4

**QUESTION 3 (10 Marks)****COMMENCE A NEW PAGE****MARKS**

(a) Given  $f(x) = \frac{x^3 - 3x^2}{x^2 - 1}$ ,

$$(x^2 - 1)(x - 3) = x^3 - 3x^2 - x + 3$$

- (i) Determine all intercepts of  $f(x)$ . 1
- (ii) Write down the equations for the horizontal and vertical asymptotes, if any. 1
- (iii) Show that the oblique asymptote is given by  $y = x - 3$ . 1
- (iv) Hence make a neat sketch of  $f(x)$ , showing all intercepts and turning points. 2

[You may assume, without proof, the following:

(i)  $f'(x) = \frac{x^4 - 3x^2 + 6x}{(x^2 - 1)^2}$

(ii)  $x^3 - 3x + 6 = 0$ , when  $x \approx -2.3$ ]

- (b) Find the area enclosed by the curve of  $y = \frac{9}{9 + x^2}$ , the  $x$ -axis and the lines  $x = 1$  and  $x = -1$ . 2
- (c) Using  $t = \tan x$ , find the general solution in radians to  $\sin 2x = \tan x$ . 3

**QUESTION 4 (10 Marks)****COMMENCE A NEW PAGE**

- (a) Using the substitution
- $u = 1 + 2x$
- , find
- 2

$$\int \frac{6 dx}{\sqrt{(1 + 2x)^3}}$$

- (b) Consider
- $y = \sqrt{3} \sin \theta - 3 \cos \theta$

- (i) Express  $y$  in the form  $R \sin(\theta - \alpha)$  where  $R > 0$ , and  $0 \leq \alpha \leq 2\pi$ . 1
- (ii) By first finding the maximum and minimum turning points, sketch a neat graph of  $y = \sqrt{3} \sin \theta - 3 \cos \theta$ , for  $0 \leq \theta \leq 2\pi$ . 3

- (c) Prove the identity:
- $2 \cos^2 \theta - 2 \cos^2 2\theta = \cos 2\theta - \cos 4\theta$
- 2

- (d) Assuming,
- $\frac{d}{dx} \left[ \tan^{-1} \frac{x}{2} + \frac{2x}{x^2 + 4} \right] = \frac{16}{(x^2 + 4)^2}$
- evaluate
- $\int_{-2}^2 \frac{dx}{(x^2 + 4)^2}$
- 2

**QUESTION 5 (10 Marks)****COMMENCE A NEW PAGE****MARKS**

(a) Prove that  $\frac{d}{dx} \left[ \tan^{-1} \frac{x}{\sqrt{1-x^2}} \right] = \frac{d}{dx} [\sin^{-1} x]$  3

(b) Evaluate  $\int_0^{\sqrt{3}} \frac{dx}{\sqrt{3-x^2}}$  2

(c) A tank which is initially empty is being filled with water at a rate given by

$$\frac{dV}{dt} = \pi \left( 100 - \frac{t^2}{100} \right) \text{ L/s}$$

Where  $V$  is the volume of water in Litres after  $t$  seconds.

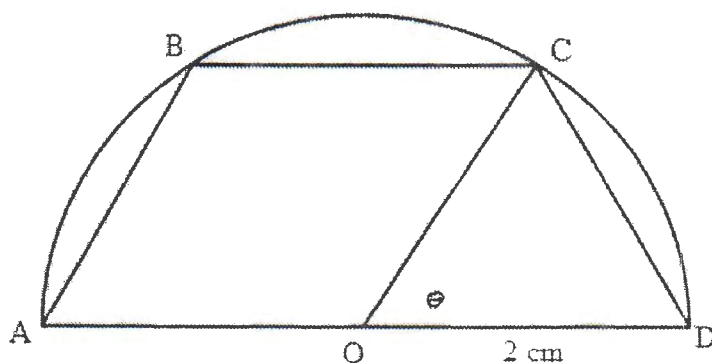
(i) At what rate is the volume of water changing when  $t = 10$  seconds? 1

(ii) How long will it take to fill the tank? 1

(iii) Find the maximum volume of water in the tank. 3

**QUESTION 6 (10 Marks)****COMMENCE A NEW PAGE**

(a) A trapezium ABCD, is inscribed in a semicircle, centre O and radius 2 cm, so that one side is along the diameter, as shown below.  $\theta$  is the angle subtended at the centre of the semicircle by one of the non-parallel sides of the trapezium.



(i) Redraw the diagram and derive an expression for the area of the trapezium in terms of  $\theta$  only. 2

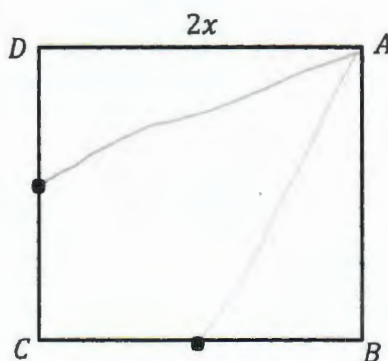
(ii) Find the maximum possible area for the trapezium. 3  
Leave answer in exact form.

- (b) If  $y = 2 \cos^{-1} \frac{x}{3}$ ,
- (i) State the domain and range of  $y$ . 1
  - (ii) Hence sketch its graph. 1
  - (iii) Express  $\cos^2 x$  in terms of  $\cos 2x$  1
  - (iv) Hence find the exact volume of the solid formed when the area bounded by the curve  $y = 2 \cos^{-1} \frac{x}{3}$ , the  $y$ -axis and the line  $y = 2\pi$ , is rotated about the  $y$ -axis. 2

**QUESTION 7 (10 Marks)**

**COMMENCE A NEW PAGE**

- (a) Find  $\int (\sin^{-1} x + \cos^{-1} x) dx$  1
- (b) Solve for  $x$ :  $\cos^{-1} x - \sin^{-1} x = \sin^{-1}(1 - x)$  2
- (c) Find the exact area bounded by the curves  $y = 2 \sin \frac{\pi x}{4}$  and  $x = 2 \sin \frac{\pi y}{4}$  for  $0 \leq y \leq 2$ . 3
- (d) (i) Prove  $\tan^{-1} \left( \frac{1}{2} \right) + \tan^{-1} \left( \frac{1}{3} \right) = \frac{1}{4} \pi$  2
- (ii)  $ABCD$  is a square field in which a goat is tethered to the corner  $A$  by means of a rope. The rope is long enough for the goat to be able to just reach the midpoints of  $BC$  and  $CD$ . Let  $DA = 2x$ .



Find the proportion of the area of the field that the goat cannot reach.

Express your answer in the form  $a + b \tan^{-1} \left( \frac{1}{3} \right)$ , where  $a$  and  $b$  are rational numbers. 2

☺ END OF EXAMINATION ☺

## MATHEMATICS Extension 1 : Question... 1...

Page 1 of 2

| Suggested Solutions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Marks | Marker's Comments                                                                                                                                                             |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>1a) <math>\int x^2 e^{x^3} dx</math></p> $= \frac{1}{3} e^{x^3} + C$                                                                                                                                                                                                                                                                                                                                                                                                                                 | 1     | No half marks                                                                                                                                                                 |
| <p>b) <math>\lim_{x \rightarrow 0} \frac{1 - \cos 2x}{x^2}</math></p> $= \lim_{x \rightarrow 0} \frac{1 - (1 - 2\sin^2 x)}{x^2}$ $= \lim_{x \rightarrow 0} \frac{2 \sin^2 x}{x^2}$ $= 2 \lim_{x \rightarrow 0} \frac{\sin x}{x} \cdot \lim_{x \rightarrow 0} \frac{\sin x}{x}$ $= 2 \times 1 \times 1$ $= 2$ <p>Alternatively: <math>\lim_{x \rightarrow 0} \frac{1 - \cos 2x}{x^2}</math></p> $= 2 \lim_{x \rightarrow 0} \frac{\sin^2 x}{x^2}$ $= 2 \times 1 \times 1 \text{ or } 2 \times 1^2$ $= 2$ | 3     | 2 marks<br>1 mark for the correct substn.                                                                                                                                     |
| <p>c) <math>\sin^{-1}\left(-\frac{\sqrt{3}}{2}\right) + \tan^{-1}\left(\tan \frac{7\pi}{8}\right)</math></p> $= -\frac{\pi}{3} + \tan^{-1}\left(\tan\left(-\frac{\pi}{8}\right)\right)$ $= -\frac{\pi}{3} + \tan^{-1}\left(-\tan \frac{\pi}{8}\right)$ $= -\frac{\pi}{3} - \frac{\pi}{8}$ $= -\frac{11\pi}{24}$                                                                                                                                                                                         | 2     | 1 mark<br>1/2 mark for $-\frac{\pi}{3}$<br>1 mark for $-\frac{\pi}{8}$<br>1/2 mark for correct answer<br>NB. $-\frac{\pi}{3} + \frac{7\pi}{8}$<br>$= \frac{13\pi}{24}$ } 1 mk |

| Suggested Solutions                                                                                                                                                                                                                                                                                                                                                                  | Marks | Marker's Comments                                                                                                 |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------------------------------------------------------------------------------------------------------------------|
| <p>1 d) (i) Prove <math>\cos(A+B) - \cos(A-B)</math><br/> <math>= -2 \sin A \sin B</math></p> <p><u>Proof</u>: LHS = <math>\cos A \cos B - \sin A \sin B</math><br/> <math>- (\cos A \cos B + \sin A \sin B)</math><br/> <math>= -2 \sin A \sin B</math></p> <p><math>\therefore</math> LHS = RHS QED</p>                                                                            | 2     | <p>1 mark to correctly expand <math>\cos(A+B)</math> &amp; 1 mark to correctly expand <math>\cos(A-B)</math>.</p> |
| <p>(ii) Hence, find <math>\cos 75^\circ - \cos 15^\circ</math><br/> <math>= \cos(45^\circ + 30^\circ) - \cos(45^\circ - 30^\circ)</math><br/> <math>= -2 \sin 45^\circ \sin 30^\circ</math><br/> <math>= -2 \times \frac{1}{\sqrt{2}} \times \frac{1}{2}</math><br/> <math>= -\frac{1}{\sqrt{2}}</math><br/> <math>= -\frac{\sqrt{2}}{2}</math> on rationalizing the denominator</p> | 2     | <p>1 mark for using the outcome in (i) correctly.</p> <p>1 mark for correct solution</p>                          |

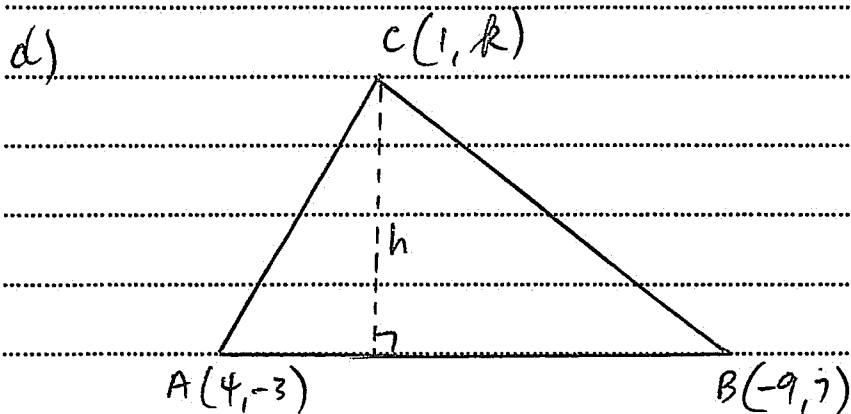
## Question 2

$$\begin{aligned} \text{a) } \frac{d}{dx}(\cos^{-1} 2x) &= \frac{-1}{\sqrt{1-4x^2}} \cdot 2 \\ &= \frac{-2}{\sqrt{1-4x^2}} \end{aligned} \quad \textcircled{1}$$

$$\begin{aligned} \text{b) } \int_0^{\frac{1}{2}} \frac{1}{2} \cos^{-1}(1-2x) dx &\approx \frac{\frac{1}{2} - 0}{6} \left[ \frac{1}{2} \cos^{-1} 0 + 4x \frac{1}{2} \cos^{-1} \frac{1}{2} + \frac{1}{2} \cos^{-1} 0 \right] \quad \textcircled{1} \\ &= \frac{1}{12} \left[ \pi_4 + 2 \times \pi_3 + \frac{1}{2} \times 0 \right] \\ &= \frac{1}{12} (\pi_4 + 2\pi_3) \\ &= 0.23998... \\ &\doteq \underline{0.24 \text{ (2dp)}} \quad \textcircled{1} \end{aligned}$$

$$\text{c) (i) RHS} = \frac{5x^2 + 20 - 16}{x^2 + 4} = \frac{5x^2 + 4}{x^2 + 4} = \text{LHS} \quad \textcircled{1}$$

$$\begin{aligned} \text{(ii) } \int_{-\frac{1}{2}}^0 \frac{5x^2 + 4}{x^2 + 4} dx &= \int_{-\frac{1}{2}}^0 \left( 5 - \frac{16}{x^2 + 4} \right) dx \\ &= 5x \Big|_{-\frac{1}{2}}^0 - 16 \times \frac{1}{2} \tan^{-1} \frac{x}{2} \Big|_{-\frac{1}{2}}^0 \quad \textcircled{1} \\ &= \frac{5}{2} - 8 \left( \tan^{-1} 0 - \tan^{-1} \left( \frac{1}{4} \right) \right) \\ &= \frac{5}{2} - 8 \left( \tan^{-1} \frac{1}{4} \right) \\ &= 0.54017... \\ &= \underline{0.54 \text{ (2dp)}} \quad \textcircled{1} \end{aligned}$$



Equation of  $AB$ :  $y+3 = \left(\frac{7+3}{-9-4}\right)(x-4)$

$$= -\frac{10}{13}(x-4)$$

$$13y + 39 = -10x + 40$$

$$\therefore 10x + 13y - 1 = 0$$

①

∴ distance from  $C$  to  $AB$ :

$$h = \frac{|10 \times 1 + 13 \times k - 1|}{\sqrt{10^2 + 13^2}}$$

$$= \frac{|9 + 13k|}{\sqrt{269}}$$

$$= \frac{|9 + 13k|}{\sqrt{269}}$$

①

$$\text{Hence } |ABC| = \frac{1}{2} \sqrt{10^2 + 13^2} \cdot \frac{|9 + 13k|}{\sqrt{269}}$$

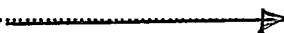
①

$$\text{i.e. } 15 = \frac{1}{2} |9 + 13k|$$

$$\therefore 9 + 13k = 30 \text{ or } -30$$

$$\Rightarrow k = \frac{21}{13} \text{ or } -3$$

①



MATHEMATICS Extension 1 : Question 3

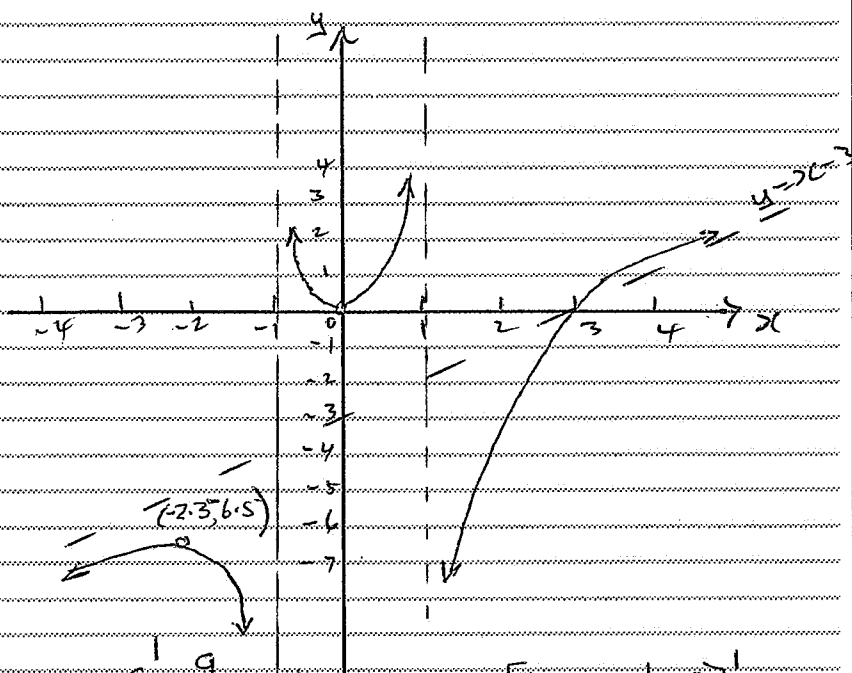
| Suggested Solutions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Marks    | Marker's Comments                                                       |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-------------------------------------------------------------------------|
| <p>a)</p> <p>(i) <math>f(x) = \frac{x^3 - 3x^2}{x^2 - 1}</math></p> <p><math>x</math> intercept <math>y = 0</math> <math>x^2(x-3) = 0</math><br/> <math>x = 0</math> or <math>x = 3</math></p> <p><math>y</math> intercept <math>x = 0</math> <math>y = 0</math>.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <p>①</p> | <p>Accept (0,0) and (3,0)</p> <p>① ① mark each.</p>                     |
| <p>(ii) No horizontal asymptote.<br/> vertical asymptotes <math>x = \pm 1</math></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <p>①</p> |                                                                         |
| <p>(iii)</p> $\begin{array}{r} x-3 \\ x^2-1 \overline{) x^3 - 3x^2} \\ \underline{x^3 - \phantom{-3x^2} + 3x} \phantom{+3} \\ -3x^2 + 3x \phantom{+3} \\ \underline{-3x^2 \phantom{+3x} + 3} \\ x-3 \end{array}$ <p><math>\therefore f(x) = (x-3) + \frac{x-3}{x^2+1}</math></p> <p>as <math>x \rightarrow \infty</math> <math>\frac{x-3}{x^2+1} \rightarrow 0</math></p> <p><math>\therefore f(x) \rightarrow x-3</math> <math>\therefore</math> oblique asymptote</p> <p>Alternatively</p> $f(x) = \frac{x^3}{x^2} - \frac{3x^2}{x^2}$ $= \frac{x^3}{x^2} - \frac{1}{x^2}$ $= \frac{x-3}{1-\frac{1}{x^2}}$ <p>as <math>x \rightarrow \infty</math> <math>\frac{1}{x^2} \rightarrow 0</math> <math>\therefore 1-\frac{1}{x^2} \rightarrow 1</math></p> <p><math>\therefore f(x) \rightarrow x-3</math> <math>\therefore</math> oblique asymptote</p> | <p>①</p> | <p>must show complete division for both methods not just cancel out</p> |
| <p>(iv) stationary point when</p> $f'(x) = 0$ $x = -2.3$ <p><math>\therefore f(-2.3) = -6.5</math></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |          | <p>No need for further calculations to be shown</p>                     |

MATHEMATICS Extension 2: Question.....

Suggested Solutions

Marks

Marker's Comments



$$b) A = \int_{-1}^1 \frac{9}{9+x^2} dx = \frac{9}{3} \left[ \tan^{-1} \frac{x}{3} \right]_{-1}^1$$

$$\text{As } \frac{9}{9+x^2} \text{ is even} = 2 \times \left[ \tan^{-1} \frac{x}{3} \right]_0^1 = 6 \tan^{-1} \left( \frac{1}{3} \right)$$

$$c) \sin 2x = \tan x$$

$$\text{Let } t = \tan x \quad x = \pi/2$$

$$\frac{2t}{1+t^2} = t$$

$$2t = t + t^3$$

$$0 = t^3 - t$$

$$0 = t(t^2 - 1)$$

$$t = 0 \text{ or } t = \pm 1$$

$$\tan x = 0 \quad x = n\pi \quad n \in \mathbb{Z}$$

$$\tan x = \pm 1 \quad x = n\pi \pm \pi/4 \quad n \in \mathbb{Z}$$

$$\text{Test } x = \pi/2$$

$$\text{LHS} = \sin \pi = 0$$

$$\text{RHS is undefined}$$

$$\therefore x = \pi/2 \text{ is not a solution}$$

②

① + ② + ③  
for each branch and appropriate asymptote

① for plotting turning point accurately and showing coordinates to scale

②

① integration

Decimal equivalent answer

1.93 accepted.

① Answer (simplified).

③

① correct substitution and simplified quadratic

② 3 quadratic solutions

① 3 general solutions

② test/reject  $x = \pi/2$

# TERM 4 MATHEMATICS EXTENSION 1 2013

## SECTION 4

a)  $\int \frac{6 dx}{\sqrt{(1+2x)^3}} = \int \frac{6}{\sqrt{u^3}} \times \frac{du}{2}$   $u = 1+2x$   
 $\frac{du}{dx} = 2$   
 $dx = \frac{du}{2}$

$= 3 \int \frac{du}{u^{\frac{3}{2}}}$  (1) integral wrt  $u$

$= 3 \left[ -2u^{-\frac{1}{2}} \right]$

$= \frac{-6}{\sqrt{1+2x}} + C$  (1) answer in terms of  $x$

(b) (i)  $\sqrt{3} \sin \theta - 3 \cos \theta = R \sin \theta \cos \alpha - R \cos \theta \sin \alpha$

$\sqrt{3} = R \cos \alpha$

$3 = R \sin \alpha$

$R^2 = 3+9$

$R = \sqrt{12}$   
 $= 2\sqrt{3}$

$\tan \alpha = \frac{3}{\sqrt{3}}$   
 $= \sqrt{3}$   
 $\alpha = \frac{\pi}{3}$

$\alpha$  is in 1<sup>st</sup> Q (1)  $R$

(1)  $\alpha$

$\therefore y = 2\sqrt{3} \sin \left( \theta - \frac{\pi}{3} \right)$

(ii)

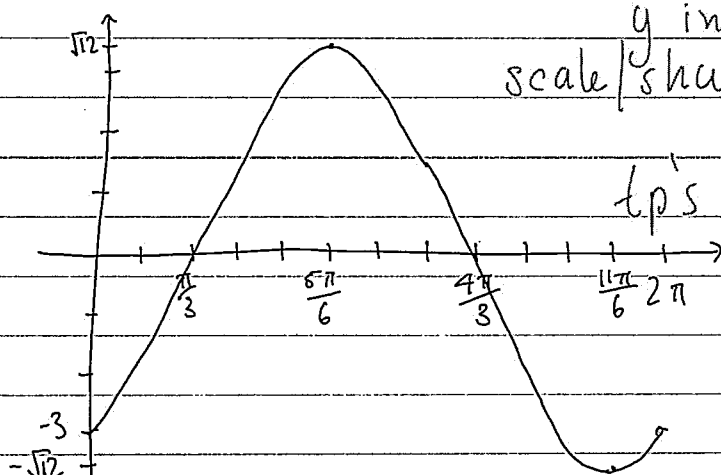
tp's are at  $\frac{\pi}{2}$  and  
shifted  $\frac{\pi}{3}$  to right

$\frac{\pi}{2} + \frac{\pi}{3} = \frac{5\pi}{6}$

$3\pi/2 + \pi/3 = \frac{11\pi}{6}$

when  $x=0$

$y = 2\sqrt{3} \sin \frac{\pi}{3}$   
 $= 2\sqrt{3} \times \frac{\sqrt{3}}{2}$   
 $= 3$



amp  
 $x$  int  
 $y$  int  
scale/shape } (1)

tp's - (1)

$$\begin{aligned}
 (c) \text{ RHS} &= \cos 2\theta - \cos 4\theta \\
 &= 2\cos^2 \theta - 1 - (2\cos^2 2\theta - 1) \\
 &= 2\cos^2 \theta - 1 - 2\cos^2 2\theta + 1 \\
 &= 2\cos^2 \theta - 2\cos^2 2\theta
 \end{aligned}$$

① double angle formulae

① answer

$$\begin{aligned}
 (d) \int_{-2}^2 \frac{dx}{(x^2+4)^2} &= \frac{1}{16} \int_{-2}^2 \frac{16}{(x^2+4)^2} dx \\
 &= \frac{1}{16} \left[ \tan^{-1} \frac{x}{2} + \frac{2x}{x^2+4} \right]_{-2}^2 \\
 &= \frac{1}{16} \left( \tan^{-1}(1) + \frac{4}{8} - \tan^{-1}(-1) - \frac{(-4)}{8} \right) \\
 &= \frac{1}{16} \left( \frac{\pi}{4} + \frac{1}{2} - \left( -\frac{\pi}{4} \right) + \frac{1}{2} \right) \\
 &= \frac{1}{16} \left( \frac{2\pi}{4} + 1 \right) \\
 &= \frac{\pi}{32} + \frac{1}{16}
 \end{aligned}$$

① primitive

no half marks

① answer

MATHEMATICS EXTENSION 1: Question 5

Suggested Solutions

Marks

Marker's Comments

$$(a) \frac{d}{dx} \frac{x}{\sqrt{1-x^2}} = \frac{\sqrt{1-x^2}(1) + x^2(1-x^2)^{-\frac{1}{2}}}{1-x^2}$$

$$= \frac{1-x^2 + x^2}{(1-x^2)^{\frac{3}{2}}}$$

$$= (1-x^2)^{-\frac{3}{2}}$$

$$\therefore \text{LHS} = \frac{d}{dx} \tan^{-1} \frac{x}{\sqrt{1-x^2}}$$

$$= \frac{(1-x^2)^{-\frac{3}{2}}}{1 + \frac{x^2}{1-x^2}}$$

$$= \frac{(1-x^2)^{-\frac{1}{2}}}{(1-x^2) + x^2}$$

$$= \frac{1}{\sqrt{1-x^2}}$$

$$= \frac{d}{dx} \sin^{-1} x$$

$$= \text{RHS as required}$$

1

Quotient rule

1

Chain rule

1

Simplification

\*Using triangles to prove  $\sin^{-1} x = \tan^{-1} \frac{x}{\sqrt{1-x^2}}$  requires great care with the resultant restrictions

$$(b) \int_0^{\sqrt{3}} \frac{dx}{\sqrt{3-x^2}} = \left[ \sin^{-1} \frac{x}{\sqrt{3}} \right]_0^{\sqrt{3}}$$

$$= \sin^{-1} 1 - \sin^{-1} 0$$

$$= \frac{\pi}{2}$$

1

Integrate

1

Evaluate boundaries

$$(c) \frac{dV}{dt} = \pi \left( 100 - \frac{t^2}{100} \right) \text{ L/s}$$

$$(i) \text{ When } t = 10,$$

$$\frac{dV}{dt} = \pi \left( 100 - \frac{100}{100} \right)$$

$$= 99\pi \text{ L/s}$$

1

Evaluate

**MATHEMATICS EXTENSION 1: Question 5**

**Suggested Solutions**

**Marks**

**Marker's Comments**

(c.) Continued.

(ii) Given no further information,  
tank is filled when  $\frac{dV}{dt} = 0$

$$\text{ie } \pi \left(100 - \frac{t^2}{100}\right) = 0$$

$$\frac{t^2}{100} = 100$$

$$t = \pm 100$$

But since  $t > 0$ ,  $t = 100$  s only

(iii)  $V = 100\pi t - \frac{\pi t^3}{300} + c$

When  $t = 0$ ,  $V = 0$   $\therefore c = 0$

When  $t = 100$ ,

$$V = 10000\pi - \frac{1000000\pi}{300}$$

$$= \frac{20000\pi}{3} \text{ L}$$

ALTERNATIVELY

$$V = \int_0^{100} \pi \left(100 - \frac{t^2}{100}\right) dt$$

$$= \left[100\pi t - \frac{\pi t^3}{300}\right]_0^{100}$$

$$= \left(10000\pi - \frac{1000000\pi}{300}\right) - 0$$

$$= \frac{20000\pi}{3} \text{ L}$$

1

Solve for  $\frac{dV}{dt} = 0$

1

Integrate

1

Constant

1

Evaluate

1

Definite integral

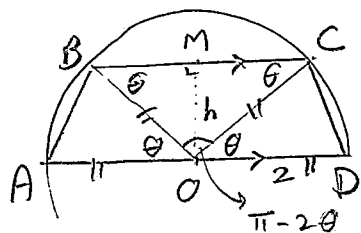
1

Integrate

1

Evaluate

(i)



$$\triangle OAB \equiv \triangle ODC \text{ (SAS)}$$

$$\text{Area } \triangle OAB = \frac{1}{2} \cdot 2 \cdot 2 \cdot \sin \theta = 2 \sin \theta$$

$$\begin{aligned} \text{Area } \triangle BOC &= \frac{1}{2} \cdot 2 \cdot 2 \cdot \sin(\pi - 2\theta) \\ &= 2 \sin 2\theta \end{aligned}$$

$$\begin{aligned} \text{Area Trapezium } ABCD &= 2 \times 2 \sin \theta + 2 \sin 2\theta \\ &= 4 \sin \theta + 2 \sin 2\theta \end{aligned}$$

$$\text{OR } h = 2 \sin \theta$$

$$BC = 2 BM = 2 \cdot 2 \cos \theta = 4 \cos \theta$$

$$\begin{aligned} |ABCD| &= \frac{1}{2} (BC + AD) \times h \\ &= \frac{1}{2} (4 \cos \theta + 4) \cdot 2 \sin \theta \\ &= 4 \cos \theta \sin \theta + 4 \sin \theta \\ \text{or } &= 2 \sin 2\theta + 4 \sin \theta \end{aligned}$$

$$\begin{aligned} \text{ii) } A' &= 4 \cos \theta + 4 \cos 2\theta \quad \text{or} \\ &= -4 \sin^2 \theta + 4 \cos^2 \theta + 4 \cos \theta \end{aligned}$$

$$\text{SP, } A' = 0 \quad \begin{aligned} 4 \cos \theta &= -4 \cos 2\theta \\ \cos \theta &= -(2 \cos^2 \theta - 1) \end{aligned}$$

$$\begin{aligned} 2 \cos^2 \theta + \cos \theta - 1 &= 0 \\ (2 \cos \theta - 1)(\cos \theta + 1) &= 0 \quad \therefore \theta = \frac{\pi}{3} \text{ or } \pi \\ \text{but } 0 < \theta < \frac{\pi}{2} \quad \therefore \theta &= \frac{\pi}{3} \text{ only} \end{aligned}$$

$$A'' = -4 \sin \theta - 8 \sin 2\theta$$

$$A''\left(\frac{\pi}{3}\right) = -2\sqrt{3} - 4\sqrt{3} = -6\sqrt{3} \doteq -10.39 < 0$$

$$\therefore \text{rel max at } \frac{\pi}{3}$$

Since  $A$  is continuous for  $0 < \theta < \frac{\pi}{2}$ ,  
 & only 1 T.P.  $\theta = \frac{\pi}{3}$  will give  
 absolute max Area

$$\text{max Area} = \frac{6\sqrt{3}}{2} = 3\sqrt{3} \text{ units}^2 \quad \#$$

no marks for diagram

1m

some do not simplify  
 $\sin(\pi - 2\theta) = \sin 2\theta$   
 $-\frac{1}{2}h$

1m

many students  
 thought the 3 triangles  
 are congruent

$\frac{1}{2}m$   
 $\frac{1}{2}m$

$$\therefore 6 \sin \theta \quad \begin{matrix} 1m \\ \text{only} \end{matrix}$$

Many students thought  
 $ABCD$  is parm

1m

1m

$$\theta = \frac{\pi}{2} \quad 0m$$

1m

Many forgot to  
 mention  $\theta \neq \pi$   
 students need to  
 mention  $\pi$  but reject  
 it later

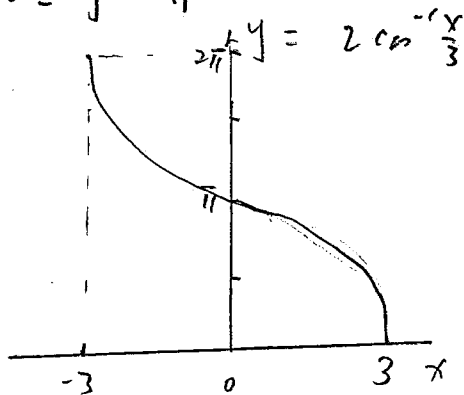
$\frac{1}{2}m$

$\frac{1}{2}m$

b)  $y = 2 \cos^{-1} \frac{x}{3}$

i) D:  $-3 \leq x \leq 3$

R:  $0 \leq y \leq 2\pi$



iii)  $\cos^2 x = \frac{1 + \cos 2x}{2}$

iv)  $\text{Vol} = \int_{\pi}^{2\pi} \pi x^2 dy = \pi \int_{\pi}^{2\pi} \left[ 3 \cos\left(\frac{y}{2}\right) \right]^2 dy$

$$\text{Vol} = 9\pi \int_{\pi}^{2\pi} \frac{1 + \cos y}{2} dy$$

$$= 9\pi \left( y + \sin y \right) \Big|_{\pi}^{2\pi}$$

$$= 9\pi \left( 2\pi + \sin 2\pi - \pi - \sin \pi \right)$$

$$= 9\pi^2 \text{ unit}^3 \#$$

$\frac{1}{2}m$

well done

$\frac{1}{2}m$

well done

1m

Pay attention  
to the vertical  
gradients at end  
points  $-\frac{1}{2}m$

1m

1m

no half mark  
some use limits  
from 0 to  $\pi$   
need to explain  
 $-1m$

1m

## Suggested Solutions

Marks

Marker's Comments

a)  $\sin^{-1} x + \cos^{-1} x = \pi/2$  (radians!!)

$$\therefore \int \sin^{-1} x + \cos^{-1} x \, dx = \int \frac{\pi}{2} \, dx$$

$$= \frac{\pi x}{2} + k$$

1

Marks were not deducted if k omitted.

b) Let  $\alpha = \cos^{-1} x$ ,  $\beta = \sin^{-1} x$

Take sine of both sides, noting that  $\cos \alpha = x$ ,  $\sin \alpha = \sqrt{1-x^2}$ ,  $0 \leq \alpha \leq \pi$   
 $\sin \beta = x$ ,  $\cos \beta = \sqrt{1-x^2}$ ,  $-\frac{\pi}{2} \leq \beta \leq \frac{\pi}{2}$

$$\sin(\alpha - \beta) = \sin \alpha \cos \beta - \cos \alpha \sin \beta$$

$$= \sqrt{1-x^2} \sqrt{1-x^2} - x \cdot x$$

$$= 1 - x^2 - x^2$$

$$= 1 - 2x^2$$

$$\sin(\sin^{-1}(1-x)) = \frac{1-x}{1-2x^2}$$

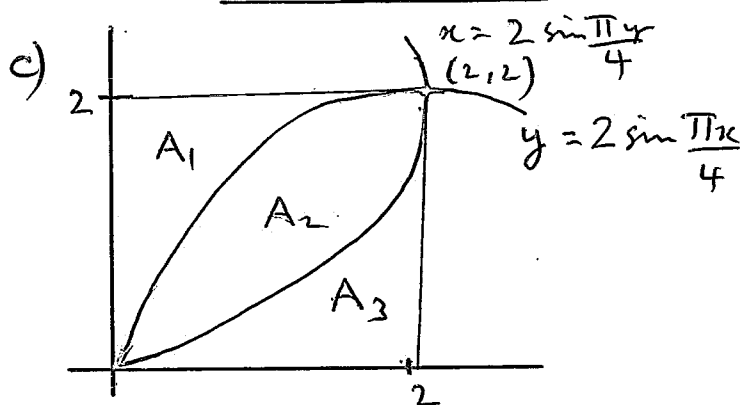
$$\therefore 1-x = 1-2x^2$$

$$2x^2 - x = 0$$

$$x = 0 \text{ or } \frac{1}{2}$$

1

1



Diagrams were scarce and generally too small and useless.

The curves are inverses of each other so  $A_1 + A_2 = A_2 + A_3$ , ( $\therefore A_1 = A_3$ ).

$$A_2 = 2(A_2 + A_3) - (A_1 + A_2 + A_3)$$

1

Mark for some sensible combination of areas.

Suggested Solutions

Marks

Marker's Comments

$$\begin{aligned} c) (\text{cont}) &= 2 \int_0^2 \sin \frac{\pi x}{4} dx - 4 \\ &= 4 \left[ -\frac{4}{\pi} \cos \frac{\pi x}{4} \right]_0^2 - 4 \\ &= 4 \left( 0 + \frac{4}{\pi} \right) - 4 \\ &= \frac{16}{\pi} - 4 \end{aligned}$$

1

Mark for a correct integral

$$\therefore \text{Common area is } \underline{\underline{\left( \frac{16}{\pi} - 4 \right) u^2}}$$

1

Mark for correct answer.

$$d) i) \left. \begin{aligned} \text{Let } \alpha &= \tan^{-1}\left(\frac{1}{2}\right), \quad 0 < \alpha < \frac{\pi}{4} \\ \beta &= \tan^{-1}\left(\frac{1}{3}\right), \quad 0 < \beta < \frac{\pi}{4} \end{aligned} \right\}^*$$

$$\begin{aligned} \therefore \tan(\alpha + \beta) &= \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta} \\ &= \frac{\frac{1}{2} + \frac{1}{3}}{1 - \frac{1}{2} \cdot \frac{1}{3}} \\ &= \frac{5/6}{5/6} = 1 \end{aligned}$$

1 Mark for answer

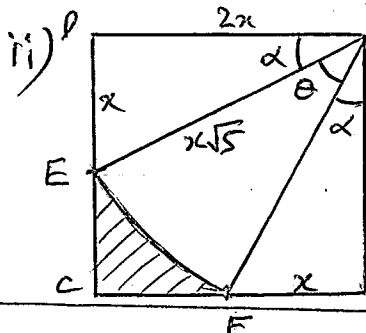
$$\therefore \alpha + \beta = \frac{\pi}{4} + n\pi \quad n \in \mathbb{Z}$$

But, from \*,  $0 < \alpha + \beta < \pi/2$

$$\therefore n = 0 \text{ and } \tan^{-1}\left(\frac{1}{2}\right) + \tan^{-1}\left(\frac{1}{3}\right) = \underline{\underline{\frac{\pi}{4}}}$$

1

1 Mark for correct handling of domains.



$$\begin{aligned} \alpha &= \tan^{-1} \frac{x}{2x} = \tan^{-1} \frac{1}{2} \\ \therefore \theta &= \frac{\pi}{2} - 2 \tan^{-1}\left(\frac{1}{2}\right) \\ &= 2 \left( \frac{\pi}{4} - \tan^{-1} \frac{1}{2} \right) \\ &= 2 \tan^{-1}\left(\frac{1}{3}\right) \text{ (part (i))} \end{aligned}$$

1/2

1/2 for  $\tan^{-1}(1/2)$

| Suggested Solutions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Marks                                    | Marker's Comments                           |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|---------------------------------------------|
| <p>d) ii) (cont) <math>EA = x\sqrt{5}</math> (Pythagoras on <math>\triangle ADE</math>)</p> <p><math>\therefore</math>, Unaccessible area</p> $= 4x^2 - \left( \frac{2}{2} \times 2x \times x + \frac{(x\sqrt{5})^2}{2} \right)$ $= 2x^2 - \frac{5x^2}{2} \left( 2 + \tan^{-1}\left(\frac{1}{3}\right) \right)$ $= \underline{2x^2 - 5x^2 \tan^{-1}\left(\frac{1}{3}\right)}$ <p><math>\therefore</math>, Proportion of total area is</p> $\frac{2x^2 - 5x^2 \tan^{-1}\left(\frac{1}{3}\right)}{4x^2}$ $= \underline{\underline{\frac{1}{2} - \frac{5}{4} \tan^{-1}\left(\frac{1}{3}\right)}}$ <p><math>(a = \frac{1}{2}, b = -\frac{5}{4})</math></p> | <p>1</p> <p><math>\frac{1}{2}</math></p> | <p>lots of people forgot this last bit.</p> |